

Conductor-Shell p-Adic Fractional Heat Semigroup: Exact Zeta, Scale Oscillation, and Boundary Closure

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Abstract

For every fixed prime p and $\alpha > 0$, we define a nonnegative operator on $L^2(\mathbb{Z}_p)$ directly as the Fourier multiplier 0 on constants and $p^{\alpha n}$ on characters of exact conductor n . Its spectrum and a conditional-expectation reconstruction prove a positive conservative Markov semigroup with an exact heat trace. We prove the exact eigenvalue staircase, a discrete-scale oscillation and log-periodic heat profile, and the sharp Schatten threshold for fractional resolvents. The fixed local prime is not promoted to all-prime target arithmetic.

1 The frozen conductor multiplier

Let p be any fixed rational prime, let $G = \mathbb{Z}_p$ have Haar mass one, and use $\widehat{G} = \mathbb{Q}_p/\mathbb{Z}_p$. A nonzero class ξ has exact conductor $n(\xi) = n$ when χ_ξ is trivial on $p^n\mathbb{Z}_p$ but not on $p^{n-1}\mathbb{Z}_p$. Exactly $p^n - p^{n-1} = (p-1)p^{n-1}$ characters have conductor n .

For $\alpha > 0$ we freeze

$$D_{p,\alpha}1 = 0, \quad D_{p,\alpha}\chi_\xi = p^{\alpha n(\xi)}\chi_\xi, \quad (1)$$

with its maximal diagonal domain. For $\mu \geq 0$, set $A_{p,\alpha,\mu} = D_{p,\alpha} + \mu I$ and $T_t = e^{-tA_{p,\alpha,\mu}}$. Equation (1) is the normalization: the paper does not identify it with every convention called a Vladimirov operator. Pseudodifferential spectra over p-adic fields, p-adic wavelet eigenbases, and ultrametric Markov semigroups form the surrounding literature [1, 2, 3, 5]. More directly, Example 5.1 of Chacón-Cortés-Zúñiga-Galindo [4], at dimension one and exponent α , already gives exactly our positive shell spectrum, multiplicities, geometric zeta, and vertical pole lattice. Those receive zero priority credit. We retain a source-local Markov, scale-boundary, determinant, and finite-reconstruction closure, not a literature-novelty claim.

Let E_n denote conditional expectation onto functions constant on cosets of $p^n\mathbb{Z}_p$, so $E_0 = P_0$ projects onto constants and $E_n \uparrow I$ strongly. Put

$$W_0 = \text{ran } E_0, \quad W_n = \text{ran } E_n \ominus \text{ran } E_{n-1} \quad (n \geq 1).$$

Then $\dim W_n = (p-1)p^{n-1}$ and the character and filtration descriptions give the same orthogonal decomposition.

Theorem 1 (spectrum and Markov reconstruction). *The spectrum of $D_{p,\alpha}$ consists of the simple eigenvalue zero and $p^{\alpha n}$ with multiplicity $(p-1)p^{n-1}$, $n \geq 1$. Its resolvent is compact. The spectrum of $A_{p,\alpha,\mu}$ is the simple eigenvalue μ and $\mu + p^{\alpha n}$ with the same multiplicities. Moreover $e^{-tD_{p,\alpha}}$ is a positive, conservative, self-adjoint contraction semigroup, while T_t is positive sub-Markov and $\|T_t\| = e^{-\mu t}$.*

Proof. Character counting and (1) prove the spectral assertion; the eigenvalues tend to infinity. A second reconstruction is

$$D_{p,\alpha} = \sum_{n \geq 1} p^{\alpha n} P_{W_n} = \sum_{n \geq 0} a_n (I - E_n), \quad (2)$$

where $a_0 = p^\alpha$ and $a_n = p^{\alpha(n+1)} - p^{\alpha n}$ for $n \geq 1$. On W_m , the right side is $\sum_{n < m} a_n = p^{\alpha m}$. For every conditional expectation E ,

$$e^{-sa(I-E)} = E + e^{-sa}(I-E)$$

is positive, constant-preserving, and contractive. The finite factors in (2) commute; their strong limit proves the Markov claims. The scalar killing factor $e^{-\mu t}$ gives the final norm and sub-Markov statement. $\square$

This proof derives positivity from the residue filtration. Nonnegative eigenvalues alone would not have established positivity preservation.

2 Heat trace, exact staircase, and Schatten boundary

For $q > 0$, write $\mathcal{S}_q$ for the operators whose singular values have finite q -sum. When $0 < q < 1$, this is the quasi-Schatten ideal with its usual quasi-norm, not a Banach Schatten class.

Theorem 2 (sharp scale theorem). *For $t > 0$, $T_t \in \mathcal{S}_q$ for every $q > 0$ and*

$$\mathrm{Tr} T_t = e^{-\mu t} \left[1 + \sum_{n \geq 1} (p-1) p^{n-1} e^{-tp^{\alpha n}} \right]. \quad (3)$$

If $N(\Lambda)$ counts the positive spectrum, then

$$p^{\alpha m} \leq \Lambda < p^{\alpha(m+1)} \implies N(\Lambda) = p^m - 1. \quad (4)$$

Consequently

$$\limsup_{\Lambda \rightarrow \infty} \frac{N(\Lambda)}{\Lambda^{1/\alpha}} = 1, \quad \liminf_{\Lambda \rightarrow \infty} \frac{N(\Lambda)}{\Lambda^{1/\alpha}} = \frac{1}{p}.$$

For $\sigma, q > 0$,

$$(I + D_{p,\alpha})^{-\sigma} \in \mathcal{S}_q \iff \alpha \sigma q > 1; \quad (5)$$

the equality case diverges.

Proof. The spectral theorem and shell multiplicities give (3), and exponential decay in $p^{\alpha n}$ dominates every geometric multiplicity; the same argument applies after taking any positive power of the singular values. Summing the first m multiplicities gives $p^m - 1$, proving (4). At a shell and just before its successor, the scaled ratios approach 1 and $1/p$, respectively. Finally, the q -sum in (5) is comparable, up to finitely many terms, to $\sum_n p^{n(1-\alpha\sigma q)}$, which converges exactly under the displayed strict inequality. $\square$

The staircase also has an exact small-time scaling profile. If $t = p^{-\alpha m} \tau$, then locally uniformly for $\tau > 0$,

$$t^{1/\alpha} (\mathrm{Tr} e^{-tD_{p,\alpha}} - 1) \xrightarrow{m \rightarrow \infty} \Phi_{p,\alpha}(\tau) = \frac{p-1}{p} \tau^{1/\alpha} \sum_{j \in \mathbb{Z}} p^j e^{-\tau p^{\alpha j}}. \quad (6)$$

The negative tail is geometric and the positive tail superexponential, so the limit follows by reindexing $j = n - m$. Reindexing once more gives $\Phi_{p,\alpha}(p^\alpha \tau) = \Phi_{p,\alpha}(\tau)$. Equations (4) and (6) record a genuine discrete-scale oscillation, not a missing smooth Weyl constant. Notice the category distinction: positive-time heat is in every $\mathcal{S}_q$, whereas the fractional resolvent has the sharp boundary (5).

References

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